Where the Headroom Lies: Block ℓ_p-Norm Statistics for Cyclic Join Bounds

Extended version: full proofs and additional experiments

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This note accompanies the ICDE 2027 submission and gives the proofs that the paper states in condensed form: the ℓ_p Composition Inequality (the validity of every LpH-CEG bound) and the Cyclic Improvement theorem, and the $bs = 1$ experiments on JOB, a TPC-H-like schema, data size and skew. Notation follows the paper.

1 The ℓ_p Composition Inequality

Setup. A hyperpath $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m)$ adds a block of relations F_j at step j, $S_j = S_{j-1} \cup F_j$, and S_m is the set of all relations. Write $A_j = \mathrm{attr}(F_j)$, $V_j = \mathrm{attr}(S_j)$ ($V_0 = \emptyset$, $V_m = V$), the interface $I_j = A_j \cap V_{j-1}$ and $E_j = A_j \setminus I_j$. Then

(F1) $E_j = V_j \setminus V_{j-1}$, so $\{E_j\}_j$ partitions V;

(F2) $I_j \subseteq V_{j-1}$.

Weights $w_{j,p} \geq 0$ ($p \in \mathcal{P} \subseteq [1, \infty]$) satisfy the attribute-wise cover constraint of the paper: for every attribute $a \in V$, bound at step $j(a)$ (i.e. $a \in E_{j(a)}$),

$$w_{j(a)} + \sum_{j' :\, a \in I_{j'}} s_{j'} \;\geq\; 1, \qquad w_j := \sum_p w_{j,p}, \quad s_j := \sum_p w_{j,p}/p \;\; (1/\infty = 0). \tag{C}$$

A *polymatroid* is $h : 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$, monotone and submodular; $h(X \mid Y) := h(X \cup Y) - h(Y)$.

Lemma 1 (Conditioning)**.** *For a polymatroid h and $Y \subseteq Z$: $h(X \mid Y) \geq h(X \mid Z)$.*

Proof. Submodularity on $X \cup Y$ and Z gives $h(X \cup Y) + h(Z) \geq h(X \cup Y \cup Z) + h((X \cup Y) \cap Z)$, and $h((X \cup Y) \cap Z) \geq h(Y)$ by monotonicity since $Y \subseteq (X \cup Y) \cap Z$. Rearrange. □

Lemma 2 (Fractional subadditivity)**.** *Let f be monotone submodular on subsets of a finite set with $f(\emptyset) = 0$, and let $\lambda_i \geq 0$ satisfy $\sum_{i:\, a \in T_i} \lambda_i \geq 1$ for every $a \in S$. Then $\sum_i \lambda_i f(T_i) \geq f(S)$.*

Proof. Order $S = (a_1, \ldots, a_k)$ and let $S_{<t} = \{a_1, \ldots, a_{t-1}\}$. By monotonicity, the chain rule and Lemma 1, $f(T_i) \geq f(T_i \cap S) = \sum_{a_t \in T_i} f(a_t \mid T_i \cap S_{<t}) \geq \sum_{a_t \in T_i} f(a_t \mid S_{<t})$. Weighting by λ_i and exchanging sums, $\sum_i \lambda_i f(T_i) \geq \sum_t f(a_t \mid S_{<t}) \sum_{i:\, a_t \in T_i} \lambda_i \geq \sum_t f(a_t \mid S_{<t}) = f(S)$, using $f(a_t \mid S_{<t}) \geq 0$. □

Lemma 3 (The obligation)**.** *For every polymatroid h and every w satisfying (C),*

$$\sum_j \Big[s_j\, h(I_j) + w_j\, h(E_j \mid I_j) \Big] \;\geq\; h(V).$$

Proof. Let $g_j = h(E_j \mid V_{j-1})$, $\delta_j = \max(0, 1 - w_j) \in [0,1]$ and $f_j(S) = h(S \mid V_{j-1})$ for $S \subseteq E_j$; f_j is monotone, submodular and $f_j(\emptyset) = 0$.

(i) For each j', the chain rule along the partition (F1) and Lemma 1 give $h(I_{j'}) = \sum_j h(I_{j'} \cap E_j \mid I_{j'} \cap V_{j-1}) \geq \sum_j f_j(I_{j'} \cap E_j)$. Hence $\sum_{j'} s_{j'} h(I_{j'}) \geq \sum_j \sum_{j'} s_{j'} f_j(I_{j'} \cap E_j)$.

(ii) Fix j. If $\delta_j > 0$ then by (C) every $a \in E_j$ has $\sum_{j' : a \in I_{j'}} s_{j'} \geq 1 - w_j = \delta_j$, so the sets $I_{j'} \cap E_j$ with weights $s_{j'}/\delta_j$ fractionally cover E_j, and Lemma 2 gives $\sum_{j'} s_{j'} f_j(I_{j'} \cap E_j) \geq \delta_j\, g_j$. If $\delta_j = 0$ the same inequality holds because the left side is non-negative.

(iii) By (F2) and Lemma 1, $h(E_j \mid I_j) \geq g_j \geq 0$, and $w_j \geq 1 - \delta_j$, so $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$.

Adding (i)–(iii): the left side is at least $\sum_j [\delta_j + (1 - \delta_j)] g_j = \sum_j h(E_j \mid V_{j-1}) = h(V)$ by the chain rule and (F1). □

Theorem 1 (ℓ_p Composition Inequality)**.** *Let D be a database, $\mathcal{Q}(D) \neq \emptyset$, and let $\mu_p(F_j \mid I_j)$ be the ℓ_p-norm of the degree sequence of E_j-values per I_j-value in a relation $R_j \supseteq \pi_{A_j} \mathcal{Q}(D)$—for instance the block join $\bowtie_{R \in F_j} R$ computed exactly on D. If w satisfies (C), then*

$$|\mathcal{Q}(D)| \;\leq\; \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}.$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$; it is a polymatroid and $h(V) = \log_2 |\mathcal{Q}(D)|$. The marginal on A_j is supported on $\pi_{A_j} \mathcal{Q}(D) \subseteq R_j$, so the ℓ_p degree inequality of Abo Khamis et al. (PODS 2024) gives, for $p \in [1, \infty]$, $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Multiply by $w_{j,p} \geq 0$, sum, and apply Lemma 3. □

Scope. The proof uses only monotonicity and submodularity, so it holds over the whole polymatroid cone and for every hyperpath, block size and $p \geq 1$. It requires (C) exactly: an LP solver may return weights violating (C) by about 10^{-7}; scaling w by $1/(1 - \varepsilon)$, for a coverage shortfall $\varepsilon = 1 - \min_a(\text{coverage of } a)$, restores (C) and raises a bound B by the factor $B^{\varepsilon/(1-\varepsilon)}$, which grows with B. At the solver's feasibility tolerance ($\varepsilon \leq 10^{-7}$) and the largest cover bound we report ($B < 2^{100}$) the factor is below $1 + 10^{-5}$; the released code applies it; the bounds reported in the paper were computed before it did and sit below their rescaled values by at most that factor. It requires exact statistics of the database being bounded (or deterministic upper bounds on them, since the bound is monotone in each μ); a sampled or sketched statistic, such as the Hutchinson estimate of the common-neighbour ℓ_2 in the paper, is outside the guarantee. When some block join is empty the query is empty and the inequality holds trivially (the released LP gives such a factor weight zero, which only loosens the bound).

Proposition 1 (Necessity)**.** *If nonnegative weights $(w_{j,p})$ make the product above an upper bound on $|\mathcal{Q}(D)|$ for every database D whose atoms' relations are independent (no two atoms are copies of one relation), they satisfy (C). Hence on a fixed hyperpath the cover LP is optimal over all weightings of its statistics that are valid on every such database.*

Proof. Fix $a \in V$ and $N \geq 2$, and let D_N give a the domain $[N]$ and every other attribute a single value, each relation holding all tuples over these domains. Then $|\mathcal{Q}(D_N)| = N$, and the block join of F_j has N tuples if $a \in A_j$ and one otherwise, so $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, N if $a \in E_j$, and

1 if $a \notin A_j$. The bound is N^{c_a} with c_a the left side of (C) at a, and $N^{c_a} \geq N$ forces $c_a \geq 1$. (The entropy of D_N's join is the modular polymatroid $h(S) = [a \in S] \log_2 N$, so the proposition also holds over the polymatroid cone.) □

Self-joins. The construction sets each atom's relation independently, which a self-join forbids: in $R(a,b) \bowtie R(b,c) \bowtie R(c,a)$ the three atoms share R, D_N is not a database of that query, and indeed weight 3 on the first step's $|R|$ and 0 elsewhere gives the valid bound $|R|^3$ although c is uncovered. Sufficiency (the Composition Inequality) is unaffected, since every self-join database is also a database of the independent query; necessity, and with it the reading that reweighting a fixed hyperpath's statistics is exhausted, holds for independent atoms only.

Why the elementary argument does not suffice. If $w_j \geq 1$ for every step, Lemma 3 follows from (iii) alone. The LP's optima on cyclic queries have $w_j < 1$ (e.g. 1/3 on the closing step of a cycle); the deficit δ_j is paid for by later interfaces, which is exactly step (ii). The cyclic improvement and the need for Lemma 2 have the same source.

Independent checks. An LP over the polymatroid cone decides Lemma 3 at fixed weights; enumerating every vertex of the feasible weight region decides it for all weights. Over 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 the minimum slack is 0 to solver precision; an adversarial search over random abstract hyperpath schemes with $|V| \leq 5$ found no violation; and over 207,000 database–block-size checks the bound never fell below the exact count (`scripts/shannon_check_wide.py`, `scripts/validity_search.py`).

2 The Cyclic Improvement theorem

Order the cycle C_k as $R_1(a_1, a_2), \ldots, R_k(a_k, a_1)$ and extend in this order with single-relation blocks. Let $N_j = |R_j|$, M_j the maximum degree and $\mu_p(R_j)$ the ℓ_p-norm of the degree sequence of R_j from a_j to a_{j+1}.

Structure. Step 1 has interface $\emptyset$ and extension $\{a_1, a_2\}$: its degree sequence is the single value N_1. A middle step $2 \leq j \leq k-1$ has interface $\{a_j\}$, extension $\{a_{j+1}\}$ and $\mu_\infty = M_j$. The closing step k has interface $\{a_k, a_1\}$ and empty extension; its conditional degrees are all 1, so $\mu_\infty = 1$ and $\mu_1 = N_k$. With $p = \infty$ and unit weights this gives $B_{\max} = N_1 \prod_{j=2}^{k-1} M_j$.

A feasible point for every common $p \geq 1$. Put weight $\frac{p}{p+1}$ with exponent p on every middle step, $\frac{p}{p+1}$ on step 1 and $\frac{1}{p+1}$ on the closing step with exponent 1. The cover constraints read: a_1: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_2: $\frac{p}{p+1} + \frac{p/(p+1)}{p} = 1$; a_i for $3 \leq i \leq k-1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_k: $\frac{p}{p+1} + \frac{1}{p+1} = 1$. All are tight, so the LP optimum satisfies

$$B_{\mathrm{LP}} \leq N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}} \quad \text{and} \quad \frac{B_{\max}}{B_{\mathrm{LP}}} \geq \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{p/(p+1)}}.$$

The exponent must be common (or non-increasing). The constraint of the attribute between middle edges j and $j{+}1$ is $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$, i.e. $p_{j+1} \leq p_j$. Choosing each edge's best exponent independently is therefore not feasible in general (e.g. $p_j = 1$, $p_{j+1} = \infty$ covers that attribute only to $\frac{1}{2}$); the LP may still mix exponents within a step.

Growth. At $p = 2$ a middle factor $M_j/\mu_2(R_j)^{2/3}$ exceeds 1 exactly when $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree. A few high-degree values satisfy it; many low-degree ones need not (n values of uniform degree d qualify iff $n < d$, and 10^4 values of degree 1 plus one of degree 20 do not). When every middle factor is at least $c > 1$ the ratio grows geometrically in the number $k - 2$ of middle edges. As $p \to \infty$ every factor tends to 1, and LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, so the realized ratio is always at least 1. The boundary factor $(N_1/N_k)^{1/(p+1)}$ does not depend on k; it is 1 for a uniform cycle.

Acyclic queries. On a star the cover constraints force unit weight on every step and the LP returns the max-degree bound. On a path the LP can use an interface at $p = 1$ in place of an extension (on a 4-path it returns weights $(1, 0, 1)$), so acyclicity alone does not force unit weights; what redundant coverage on cycles adds is the slack exploited above.

3 A spectral reading of the common-neighbour statistic (post hoc)

For an even k-cycle of binary relations the count is a trace, $|\mathcal{Q}| = \mathrm{Tr}(A_1 \cdots A_k)$ for the incidence matrices A_i ($v_i \times v_{i+1}$), and the common-neighbour ℓ_2 norm is the Frobenius (Schatten-2) norm of the block product $B = A_i A_{i+1}$. Hölder's inequality for Schatten norms gives, over $m = k/2$ consecutive blocks, $|\mathcal{Q}| \le \prod_j \|B_j\|_{S_m}$, and $\|B\|_{S_m}^m \le \sigma_{\max}(B)^{m-2}\|B\|_F^2$ with $\sigma_{\max}$ bounded above by a Collatz–Wielandt certificate on $B^\top B$. Two numbers per block, both from the pair join ($\sigma_{\max}$ lies outside the polymatroid catalog), give a bound tighter than the polymatroid program on 96/240 long-suite queries (by a median 1.25× there, though 1.11× looser over all) and never below an exact count; with each block's top 24 singular values (Lanczos; uncertified, yet above every exact count) it is tighter on 220/240 and takes IMDb at $|V|$=10 from 14.9× to 3.2× the truth. The next step on the statistics axis may thus be spectral order rather than block size.

4 Which statistics are new: a vocabulary table and a separation

For a binary block $R(a, b) \bowtie S(b, c)$ the statistics the paper uses or compares against are the following ($\deg_R(b)$ is the number of a with $(a, b) \in R$, and $N_R(a)$ the set of b with $(a, b) \in R$).

Statistic	Definition	Pair join?	Used by
single-relation norms	$\|\deg_R(\cdot \mid a)\|_p$, $\|\deg_R(\cdot \mid b)\|_p$	no	LpBound, all programs
pair-join size	$\sum_b \deg_R(b)\deg_S(b)$	no	CorrBound (ℓ_1 inner product)
shared-attribute products	$\sum_b \deg_R(b)^p \deg_S(b)^q$	no	CorrBound's generalized inner products (per its authors)
one-side interface norms	$\big\|\big(\sum_{b\in N_R(a)} \deg_S(b)\big)_a\big\|_p$	no	this paper (join-free)
common-neighbour norms	$\big\|\big(\lvert N_R(a) \cap N_S(c)\rvert\big)_{a,c}\big\|_p$	yes	this paper

Proposition 2 (Separation)**.** *The common-neighbour ℓ_2 norm is not determined by the other rows of the table: there are two databases that agree on every single-relation degree sequence, on the pair-join size, on every shared-attribute product $\sum_b \deg_R(b)^p \deg_S(b)^q$ ($p, q \ge 0$) and on every one-side interface norm, yet differ in the squared common-neighbour ℓ_2 norm (32 against 24), and so in the number of 4-cycles they close.*

Proof. Take R with the 4×4 incidence matrix M and S with $M^\top$, for $M = A$, two disjoint copies of $K_{2,2}$, and $M = B$, the bipartite 8-cycle (row i adjacent to columns i and $i{+}1 \bmod 4$).

Both are 2-regular, so every degree in R and S is 2, every shared-attribute product is $4 \cdot 2^{p+q}$, the pair-join size is 16, and every one-side interface degree is 4. The common-neighbour matrix is $MM^\top$: for A it has eight entries equal to 2 (squared norm 32); for B, four diagonal entries 2 and eight entries 1 (squared norm 24); the largest common-neighbour count is 2 in both. For the 4-cycle $R(a,b) \bowtie S(b,c) \bowtie R'(c,d) \bowtie S'(d,a)$ with $R' = R$, $S' = S$ as separate atoms, the count is $\mathrm{tr}((MM^\top)^2) = \|MM^\top\|_F^2$, again 32 against 24. □

The separation is limited to the statistics listed: it does not claim that no multi-relation term of CorrBound, or any other statistic, determines the common-neighbour norm, and it says nothing about the size of the gain on real data (the paper's ablations measure that).

5 Which statistic needs the join: three-norm details, sketch and selection

Of the eight interfaces of a block of binary relations $R(a,b), S(b,c)$, seven (five non-trivial) come from a scan of one relation with degree lookups in the other (the degree of a is $\sum_{b \in N_R(a)} \deg_S(b)$, that of b is $\deg_R(b) \deg_S(b)$); only $\{a,c\}$, the common-neighbour count $|N_R(a) \cap N_S(c)|$, needs the pair join; its ℓ_2 is determined neither by the other interfaces nor by shared-attribute inner products (Section on the separating example), while its ℓ_1 is the join size. We compute the seven without the join (identical to the materialized statistics on 120/120 Hetionet queries; 0.49 s against 4.6 s per block). With three norms and the flow LP they stop a median 2.3× (max 18.7×) above the full bs=2 bound on the Hetionet long suite (1.5–2.9× on the families, 1.9–57× on the sweep), and the common-neighbour ℓ_2, $\|A_R^\top A_S\|_F$, closes the gap on 12/12 family–seed pairs, 10/10 sweep points, 85/120 Hetionet and 116/120 IMDb queries (Hetionet median 1.00×, max 1.67×); its ℓ_∞ does not (1.70×), and added to bs=1 alone it reaches 1.03× (53/120 exact), so the attribution is order-dependent. In the headline setting (eleven norms, polymatroid) the join-free seven carry a median 88% and 86% of the log-gain on Hetionet and IMDb, the ℓ_2 leaves 1.47× and 1.39× (full on 1/120 each), and the other common-neighbour norms (with the join-free ℓ_1) close the rest. A one-pass triangle-inequality bound on the ℓ_2 recovers nothing on Hetionet (2.2×); a 64-probe Hutchinson sketch lands within 0.91–1.78× of the full bound in 0.32 s per block but is an estimate, not a bound. Post hoc, it picks which joins to pay for: joining the three of a $|V|$=6 query's six blocks whose sketched norm lowers the bound most recovers the full $bs \leq 2$ bound on the median of 34 Hetionet queries (top two on 32 IMDb ones), against half that gain for a random three; timed end to end, this plan takes a median 0.67× the time of joining every block on Hetionet (33/40 cheaper) but 1.39× on IMDb, where join-free statistics, sketches and ranking cost more than the three joins they skip. These times are against materialized pair joins. The sparse-product path of the paper's Table V computes every common-neighbour norm without join tuples, for 8.2× and 8.3× LpBound's collection, so choosing which blocks to join matters only where the joins are materialized.

6 Which block size to pay for: cost-study details

At bs=1, collecting degree sequences on the four cost-study families costs 22–27 ms against ∼1.9 ms to solve; $bs \geq 2$ multiplies that term by a median 45× there (24× more entries) and 61× (8–558×) on the Hetionet suite, and a cataloged bs=2 estimate costs 10 ms (the paper's long-suite table). Use bs=1 when the bound is re-solved many times per statistics build and $bs \geq 2$ when the pair joins can be materialized ahead of the workload.

7 IMDb at five times the films, and the YAGO count gap

IMDb at 1,000 *films.* At five times the films of the paper's slice (1,000; the 40 queries at $|V|$=6, bounds only, three norms; exploratory) every tightness ratio grows: the inference gap from 3.8 to 4.8×, block statistics from 4.0 to 7.0× under the cover LP and from 17 to 48× under the flow LP, 40/40 below entropic, the program gap from 4.2 to 6.8×. Collecting the statistics grows faster than the data: 4.2× the rows cost 48 s per query at the median (from 4.4 s), up to 25 min and 16 GiB, measured with 10 workers sharing the machine.

YAGO counts. Our exact counts differ from G-CARE's on 49 of the 192 YAGO queries we evaluate. A second implementation of the same semantics that reads G-CARE's raw files without DuckDB reproduces our count on all 49 of them, and 45 of them lie in G-CARE's Petal families (45 of 100 Petal queries, against 4 of 92 others), so the gap is in semantics or preprocessing we could not reconstruct, not settled either way.

8 Degree sequences from catalog histograms

Degree sequences can come from catalog histograms at `ANALYZE` time: over buckets (d_i, c_i) (degree d_i held by c_i values), $\|\mathbf{d}\|_p = (\sum_i c_i d_i^p)^{1/p}$. We have not evaluated this.

9 CorrBound's cyclic workload

CorrBound's public artifact (`https://github.com/christophmayerm/CorrBound`, MIT) ships the workload of its cyclic experiment: 400 labelled subgraph-matching queries with 8 vertices (200 dense, 200 sparse; 11–28 edges) on DBLP (317K vertices, 2.1M directed edge rows, symmetric), its estimates for CorrBound-e (exact degree vectors) and LpBound (flow bound), and true cardinalities. Each edge atom becomes a binary relation over its two query vertices holding the edges whose endpoints carry the query's labels; the vertex atoms are key lookups with a label predicate, and CorrBound's own LP drops them too. CorrBound-e's and the artifact LpBound's estimates are taken from the artifact's result files, not recomputed.

Same norms, same baseline. CorrBound's LpBound statistics use $p \in \{1, \ldots, 30, \infty\}$, and so do we. Our single-relation polymatroid bound reproduces the artifact's LpBound estimate on 400/400 queries (ratio 1.000000 at both extremes; the artifact computes it with the flow LP and we with the polymatroid program, which coincide for single-relation degree constraints), so both sides start from the same statistics and the same baseline.

Results. Block statistics ($bs \leq 2$, every interface) under the polymatroid program sit a median 1.99× above the exact count (P95 3.9×), against 6.1× (P95 20×) for CorrBound-e and 64.6× (P95 190×) for LpBound. They are tighter than CorrBound-e on 382/400 queries (191 of each kind), a median 3.0×, and at most 1.51× looser on the other 18. No bound of any method falls below the exact count.

The counts. The artifact's true cardinalities sit below the count its own SQL returns on its own data on all 400 queries (a median 0.6%, at most 39%); we checked three queries by running the artifact's SQL verbatim, and requiring distinct images for same-label vertices does not reproduce them. We therefore check validity against a recount with DuckDB over the same relations; against the artifact's counts no bound falls below either.

What each side has. CorrBound-e adds, for every query vertex with k incident edges, the k-way generalized inner products of their degree vectors ($p = 1, 2, 3$) and intra-relation correlation norms; our $bs \leq 2$ catalog has neither. It has the interface-conditioned norms of every pair block, which

CorrBound's vocabulary lacks for the one-side and common-neighbour interfaces (Proposition 2). The two are complementary, and the 18 queries where CorrBound-e is tighter suggest that combining them would help. CorrBound's statistics are computed once per dataset; ours cost a median 25 s per query in our Python prototype (all pair blocks, 31 norms), and the solve 0.38 s.

10 Condensed in the main text

Three passages are shortened in the paper. *Pair-join sizes on IMDb* (the paper's IMDb paragraph keeps the self-join explanation): where half the pairs are self-joins, sizes add almost nothing over LpBound, while the interface-conditioned statistics add 17–171×. *Collection cost:* the 61× overhead of blocks, growing faster than the data, calls for cataloging inside the system rather than per-query collection. *Join evaluation:* Yannakakis+ makes Yannakakis-style evaluation of acyclic queries practical in DuckDB and PostgreSQL, the complement of our cyclic setting; CompactLTJ shrinks Leapfrog Triejoin; Capelli et al.'s sampling variant takes tree-superadditive prefix upper bounds, which our residuals could supply if they qualify, though in our 66-query study they do not beat max-degree ordering in general.

11 Intermediate norms on an acyclic path

Theorem 4 (cyclic improvement) is a sufficient condition. Intermediate norms can also tighten the cover bound on a path. Take $Q = R(a,b) \bowtie S(b,c) \bowtie T(c,d)$ with $R = \{(a_0, b_0)\}$, $S = \{(b_0, c_i) : 0 \le i < 10\}$ and $T = \{(c_0, d_j) : 0 \le j < 100\} \cup \{(c_i, d_0) : 1 \le i \le 900\}$, so $|R|, |S|, |T| = 1, 10, 1{,}000$ and $|Q| = 109$. On the hyperpath R, S, T with bs=1, the cover LP gives 1,000 with $\mathcal{P} = \{1, \infty\}$ and $\sqrt{10} \cdot \sqrt{10{,}900} \approx 330.15$ with $\mathcal{P} = \{1, 2, \infty\}$: weights $(1, \frac{1}{2}, 1)$ with exponents $(1, 1, 2)$ cover the attributes (a, b, c, d) at $(1, 1.5, 1, 1)$, and the ℓ_2 of T's c-degrees ($\sqrt{100^2 + 900}$) replaces part of S's extension. The cardinalities are the same on both sides; only ℓ_2 is added. A second instance (with $|S| = 22$, $|T| = 100$) goes from 100 to $\sqrt{20 \cdot 190} \approx 61.64$. Both values are what the released estimator returns, and LpBound's entropic bound on the same statistics equals them. Our experiments do not measure how often this happens on acyclic workloads, and the stars we ran gain nothing (each leaf is covered by one step only).

12 G-CARE YAGO by topology

The YAGO set we evaluate is the part of G-CARE's cyclic queries that fits the prototype's size caps, so it is not a random subset, and our counts differ from G-CARE's on part of it. Table 1 breaks it down (`scripts/gcare_topology.py`); gain is LpBound's entropic bound over the polymatroid bound on block statistics, eleven norms. The 143 agreeing queries are the external check proper; the 49 others are our reconstruction of G-CARE's semantics. Inclusion depends on topology (Cycle_6: 25 of 90; Petal_6: 50 of 78), and so does the gain, so the overall median describes this subset, not YAGO's cyclic workload. On AIDS all 241 queries run and agree (median 15.2×).

13 Bootstrap intervals of the gains

The paper reports that query-level bootstrap intervals keep each median ratio to LpBound on one side of 1. For the polymatroid program on block statistics that side is guaranteed by construction (its constraints contain LpBound's), so the informative quantity is the size of the gain. Table 2

Table 1: YAGO cyclic queries by topology: G-CARE's set, evaluated, count equal to G-CARE's, and median gain (all / agreeing / not).

Topology	cyclic	run	agree	gain	agree	not
Clique_10	8	8	7	2.84×	2.84×	71.2×
Clique_6	1	1	1	1.54×	1.54×	–
Cycle_3	57	41	41	2.92×	2.92×	–
Cycle_6	90	25	25	68.9×	68.9×	–
Cycle_9	12	3	3	5,930×	5,930×	–
Flower_6	12	10	10	8.67×	8.67×	–
Graph_12	2	0	0	–	–	–
Graph_3	2	2	0	168×	–	168×
Graph_6	10	2	1	161×	316×	6.69×
Graph_9	10	0	0	–	–	–
Petal_12	22	13	1	1,213×	1,213×	1,427×
Petal_6	78	50	50	31.3×	31.3×	–
Petal_9	62	37	4	87.5×	39.0×	93.2×
all	366	192	143	24.4×	12.1×	311×

gives it: median gain over LpBound (eleven norms) with its 95% interval, from 10,000 resamples of the registered queries of the fixed databases (`results/suite/bootstrap_ci.csv`). The intervals cover query sampling only, not other data snapshots, row samples or the pilot choices.

Table 2: Median gain over LpBound with 95% bootstrap interval.

Suite	$\|V\|$	cover $bs{\le}2$	interval	polymatroid	interval
Hetionet	6	45.7×	[31.1, 69.3]	236×	[155, 335]
Hetionet	8	269×	[181, 337]	1,724×	[1,213, 2,567]
Hetionet	10	920×	[767, 1,204]	10,160×	[7,831, 13,077]
IMDb	6	1.27×	[1.14, 1.34]	9.73×	[8.25, 12.2]
IMDb	8	2.88×	[2.58, 3.33]	29.8×	[24.8, 34.5]
IMDb	10	6.83×	[5.97, 7.94]	86.8×	[71.1, 96.5]

14 Statistics outside a proxy for CorrBound's

The interfaces of a block of two binary relations $R(a,b), S(b,c)$ fall into four groups by what they count: the pair-join size ($\emptyset$); functions of $\deg_R(b)$ and $\deg_S(b)$ alone ($\{b\}, \{a,b\}, \{b,c\}, \{a,b,c\}$), whose ℓ_p norms are the inner products $\sum_b \deg_R(b)^q \deg_S(b)^r$ with $(q,r) \in \{(1,1),(p,p),(1,p),(p,1)\}$, a subset of CorrBound's (its Lemma 3.1 allows a separate p_i per relation and k-way products); the one-side interfaces $\{a\}, \{c\}$; and the common-neighbour interface $\{a,c\}$. Table 3 adds them in that order to LpBound's single-relation statistics (eleven norms, polymatroid program, median gain over LpBound over 120 queries per suite; `scripts/ablation_overlap.py`). The proxy is exact (no sketches) but covers only these exponent pairs and no k-way products, so the shares below are relative to it. Beyond it, the one-side and common-neighbour statistics tighten the bound on 120/120 Hetionet and 120/120 IMDb queries, a median 34.4× and 13.9×, and carry a median 48% and 79% of the log-gain; on their own they come within 1.03× and 1.03× of the full catalog's median.

Table 3: Cumulative catalogs: median gain over LpBound.

Catalog	Hetionet	IMDb
LpBound (single relation)	1×	1×
+ pair sizes	15.1×	1.00×
+ every pairwise product	41.8×	1.97×
+ one-side	608×	18.4×
+ common-neighbour (full)	1,724×	29.4×
one-side + common-nbr only	1,674×	28.7×

15 Tail robustness by shape, with every timing kept

`scripts/robustness_repeat_raw.py` repeats the paper's 24-query execution study with every observation written out (`results/e2e/robustness_raw.csv`: query, plan, run, position in the shuffled run, time). Each plan runs six times in shuffled order after a warm-up; the reference plan (fastest forced left-deep order) is chosen on runs 1–2 and measured on runs 3–6, so choosing and measuring do not share a run. Table 4 gives the slowdowns over that reference and the paired ratio of the bound-guided plan's time to DuckDB's. The pooled P95 is 1.11× for the bound-guided order against 5.26× for DuckDB. The tail comes from the denser shapes: DuckDB is faster on triangles and 4-cycles and slower on diamonds and K_4, and the AGM filter that sets the suite depends on shape and size, so the pooled percentiles describe these four shapes, not a workload.

Table 4: Slowdown over the reference (median/max, evaluation runs) and median bound-guided/DuckDB time ratio, by shape.

		bound-guided		DuckDB		ratio
Shape	n	med	max	med	max	
triangle	6	0.83	1.12	0.78	0.94	1.13
4-cycle	6	1.02	1.10	0.59	0.70	1.78
diamond	6	1.04	1.75	2.39	2.81	0.53
K_4	6	1.00	1.05	4.25	5.92	0.25
all	24	1.00	1.75	0.88	5.92	1.01

16 Cost measurement details

The script `cost_accuracy_indep.py` behind the paper's Table V runs every (query, catalog) pair as its own task in a fresh process (8 workers, shuffled order, one run), times collection (data loading excluded), the constraint build and the solve separately, and reads the process's peak RSS after loading and at the end; their difference is a lower bound on the extra memory. The sparse-product and materialized full catalogs give the same bound on all 240 queries (largest $|\Delta \log_2|$ 0). Peak memory over the loaded data is a median 60 against 626 MB on Hetionet and 44 against 262 MB on IMDb; the largest process peaks are 429 against 6,351 MB and 341 against 4,149 MB. Building the constraints takes about 2 ms. An earlier version of the table measured the catalogs of a query in one process, in a fixed order, with the full catalog's joins materialized (`cost_accuracy.py`); its join-free and ℓ_2 ratios (6.0/6.9× and 6.2/7.1×) are close to these, and its full-catalog ratio (52/24×) is the materialized path's.

17 Unified comparison on synthetic cycles

Table 5: Unified comparison of guaranteed upper bounds (synthetic power-law cycles, 100V/400E, one instance). "Cover" is our bs=1 LP; "Entropic" is LpBound's bound (via the $2^{|V|}$ program). Lower is tighter. "/CEG" is CEG_M's ratio to Cover and "/Ent" Cover's ratio to the entropic optimum.

Query	AGM	CEG_M	Cover	Entropic	/CEG	/Ent
Cycle-3	8.0K	15.6K	6.8K	5.0K	2.3×	1.36×
Cycle-4	160K	655K	116K	87K	5.7×	1.32×
Cycle-5	3.2M	25.5M	1.97M	1.45M	13.0×	1.36×
Cycle-6	64M	1.12B	31.5M	25.2M	35.7×	1.25×

18 Additional $bs = 1$ experiments: JOB, TPC-H-like, data size and skew

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at $bs = 1$ with weights w=0.5, and a star query by exactly 1.00×; the IMDb suite of the paper is the systematic version. On a small TPC-H-like synthetic schema (SF=0.01; not `dbgen` data) every cyclic query we built yields 1.00×; LpH-CEG needs *cyclic structure* together with *degree skew* in a middle relation. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at α=0.5, 16.1±0.8× at α=2.0, 6.6±0.4× at α=3.0. At low skew all ℓ_p-norms nearly coincide, at extreme skew ℓ_∞ is already near-tight, and in between ℓ_2 captures the heavy tail.

Every number in this section is produced by the scripts in the repository and checked by its claim audit.

19 Multi-way ordering by per-step residuals

The per-step residuals can order a worst-case-optimal join greedily. As implemented, the residual of a variable prefix is the LP re-solved on the query with the prefix attributes projected out, which bounds the residual for every value of the prefix (and could be computed for LpBound in the same way): GenericJoin iterations drop 2.3× on the Hetionet cycle-4 in our simulator, but over 66 synthetic queries the bound-guided order is 1.3% worse at the median (20 wins, 43 losses; Wilcoxon p=0.003, indicative, as queries share shapes and seeds), so it is a targeted heuristic.

20 Complexity

Collecting $\mu_p(F \mid I)$ for a single relation is one scan, $O(N)$; for a two-relation block it materializes the pair join, except for the interfaces computable from a scan of one relation with degree lookups in the other (all but the common-neighbour interface of a binary block; paragraph "Which statistic needs the join" of the paper). The cover LP has $m|\mathcal{P}|$ variables and $|V|$ constraints; it solves in milliseconds for the queries in the paper. The flow LP over a $bs \leq 2$ catalog is polynomial in the number of constraints and solves in 0.03–4.4 s at k = 3–16 over $4k$ orders.